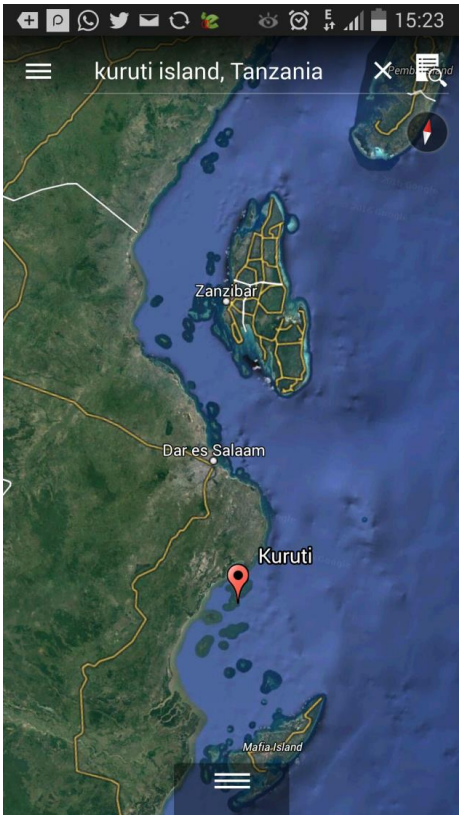
RESCO Installs a Solar Mini-Grid in Tanzania

Introduction

RESCO (T) Limited, a solar energy contractor and African Energy customer has installed an 11 KWp Solar mini-grid in Kuruti village, a small Tanzanian island located in Indian Ocean. The mini-grid serves 66 customers including households, small businesses, a school and a clinic in the island with an estimated population of 1213 and 192 households. Main applications of the power include lighting, phone charging, refrigeration of soft drinks and running Televisions.



Location of Kuruti Island in Tanzania

Power Generation

The mini-grid is DC coupled with a PV array consisting of 36 pieces of 305 Wp Suntech solar modules, four 60A outback MPPT Charge controllers, 7 KW Outback Radian inverter and a 144 KWh OPZs battery bank.

11KWp Mini-Gird Array in Kuruti Island

Outback inverter and charge controllers in the Power House in Kuruti Island

Power Distribution

The mini-grid has so far connected 34% of the island population through a 2.6 km, low voltage distribution network. The network has an outreach to 50% of the households in the island. Power is supplied to the customers via 50 mm^2 ABC cables between

the poles and a 25 mm^2 ABC service drop cable.

Construction of Power Distribution Line in Kuruti Island

Customer Connection and metering

Customers are connected through smart meters whereby electricity supply and uses will be monitored and controlled through a web-based application provided by the meter supplier. The metering has features for load limiting and prepayment billing. Customers have an option to pay bills through their mobile phones.

Left: Connection to a customer's house in Kuruti Island
Right: Customer switching on a lamp in a restaurant for the first time.

Tariffs

The tariffs are set according to the customer's ability and willingness to pay for electricity. A preliminary survey in the island indicated that a household spends between USD 7.9 to USD10.6 on energy for lighting, phone charging and power for radio per month. This is equivalent to 7% of a typical household monthly income in the island. An average household is expected to consume 0.34 KWh a day for these energy applications and the tariff is set at USD 0.44 per KWh. Customers are encouraged to use LED lamps to get the most benefits out of the services.

Challenges

Customers were not able to afford wiring costs in their houses estimated at USd130 for a simple wiring. The project organized procurement of ready boards, a pre-wired unit with a main switch, ac breakers, three socket outlets and one LED lamp. The unit costs about USD 30 at subsidized rate where the market price is estimated at USD 50. With ready boards, customers are able enjoy the electricity services while they are waiting for extension of cables to the other more appliances in their houses as their incomes allow.

An example of ready board installed at customers house in Kuruti Island

After installation of the mini-grid, the demand for more energy services has raised. More energy services are demanded for fans, TV and refrigerators. Very few households had these appliances before and the enthusiasm is more for the services.

Future Plans

RESCO plans an extension of the plant to at least 50 KWp in size in order to meet most of the energy demand in the island. That will be able to serve the customers who already have access to the existing power distribution network. An additional 2 km of distribution line will also be constructed to reach all other remaining households in the island.